break down food molecules into simpler molecules. For example, amylase coverts starch into maltose and pepsin breaks proteins into smaller peptides.

4.1. Nature and Properties of Enzymes:

1. Enzymes are proteins.
2. Enzymes are complex macromolecules with high molecular weight.
3. Enzymes acts as catalysts for biochemical reactions and are called biocatalysts.
4. They help in the breakdown of larger molecules into smaller molecules (catabolism) and to the synthesis larger molecule (anabolism).
5. Enzymes possess high degree of specificity.
6. Most of the enzymes have high turnover number.
7. Some enzymes possess only one polypeptide chain. They are known as monomeric enzymes. (Ribonuclease, Trypsin etc.)
8. Some enzymes possess more than one polypeptide chain. They are known as oligomeric enzymes. (Lactate dehydrogenase (LDH)).
9. The activity of an enzyme increases with increase in substrate concentration and ultimately reaches a steady maximum velocity.
10. Some enzymes can possess multiple function and more number of polypeptide chain are known as multi-enzyme complexes. Example: Fatty acid synthase.
11. Each enzyme show high activity at a particular pH and temperature called as its optimum pH and temperature respectively.

Note

Turnover number of an enzyme is defined as the maximum number of substrate molecules which can be converted to products per molecule of enzyme per unit time.

4.2 Nomenclature and Classification of Enzymes:

In early days the suffix – *ase* was added to the substrate for naming the enzymes.

Example : *Lipase* acts on lipids.

These names are known as trivial names. They do not convey complete information about the enzymatic reaction.

The International Union of Biochemistry and Molecular Biology (IUBMB) have assigned a systematic nomenclature for enzymes. The systematic name has two parts.

- The first part represents the substrate. In enzyme catalyzed reactions the reactants are known as substrates.

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